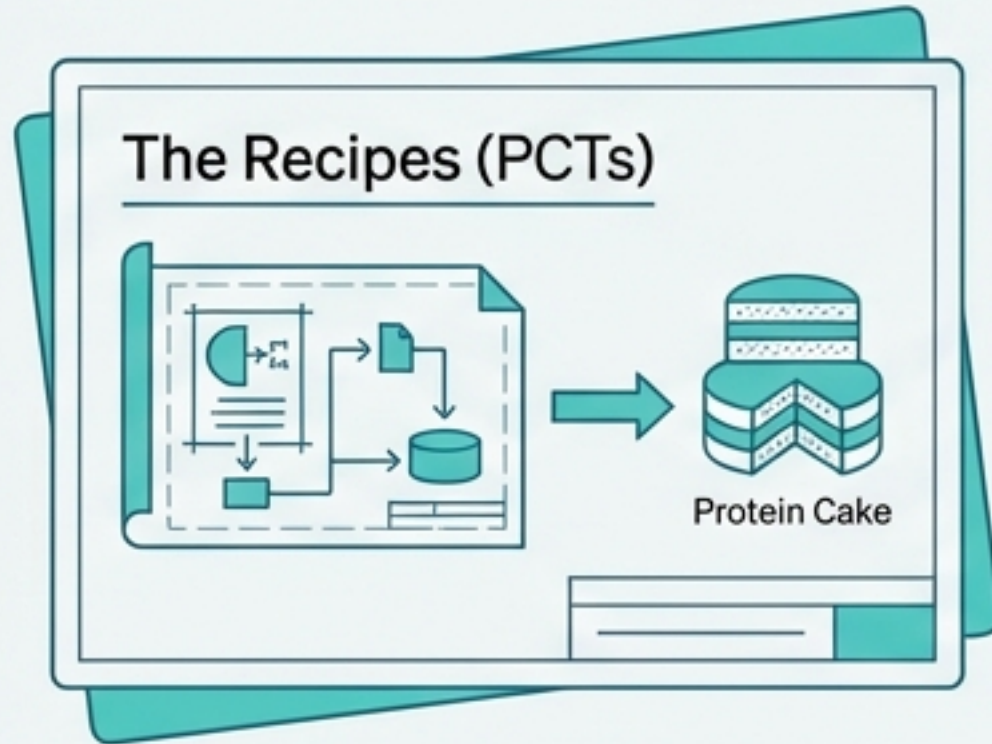


Decoding the Hidden Managers of Our DNA

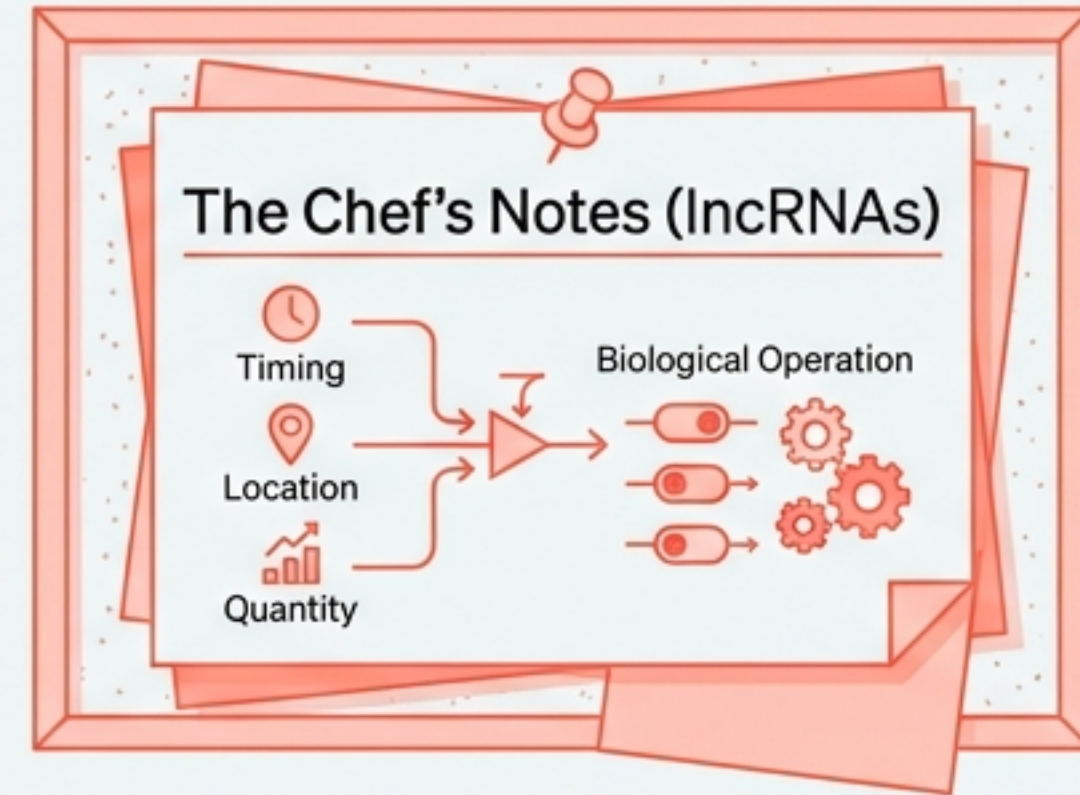


How metaLncRNA uses an ensemble of AI experts to find the genome's most elusive signals.

The Genome is a Kitchen. We've Been Ignoring the Chef.



Standard instructions that tell the cell exactly how to build a protein cake. Highly structured and easy to read.



Crucial regulatory instructions that dictate when, where, and how much to bake. They manage the entire biological operation.

Identifying these Chef's Notes is vital for understanding gene silencing and human disease.

The Camouflage Problem

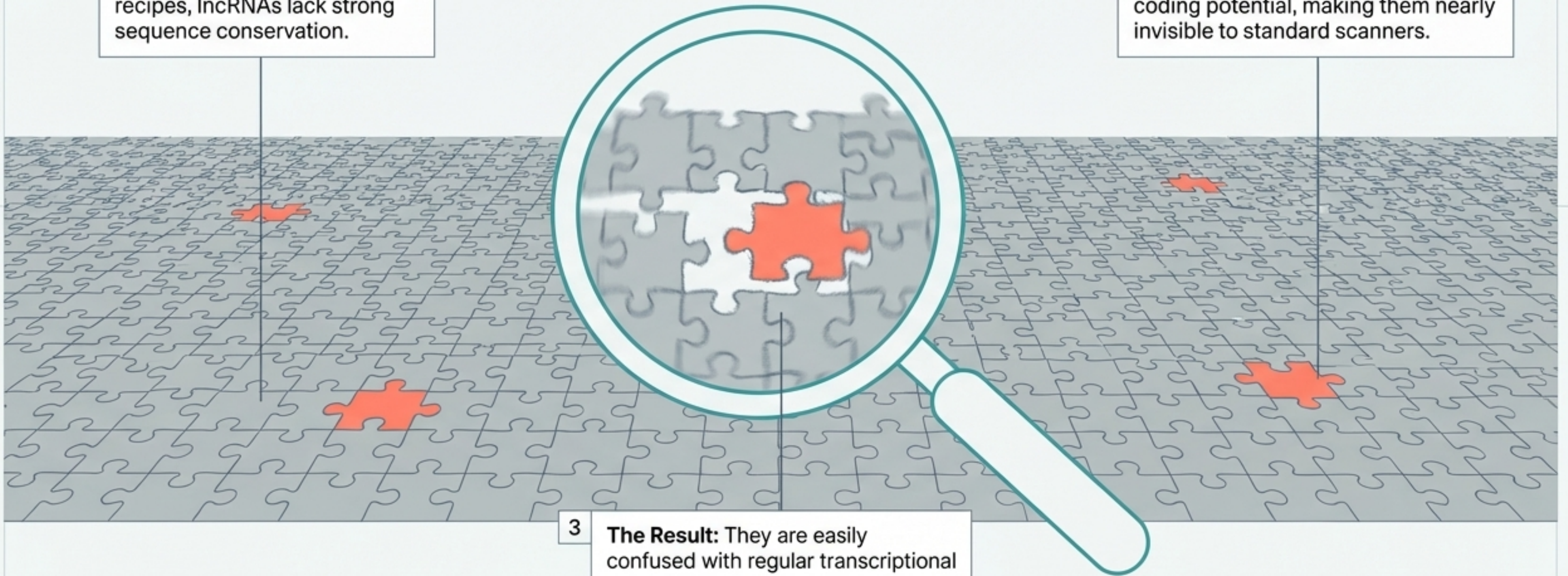
1

No Standard Structure: Unlike recipes, lncRNAs lack strong sequence conservation.

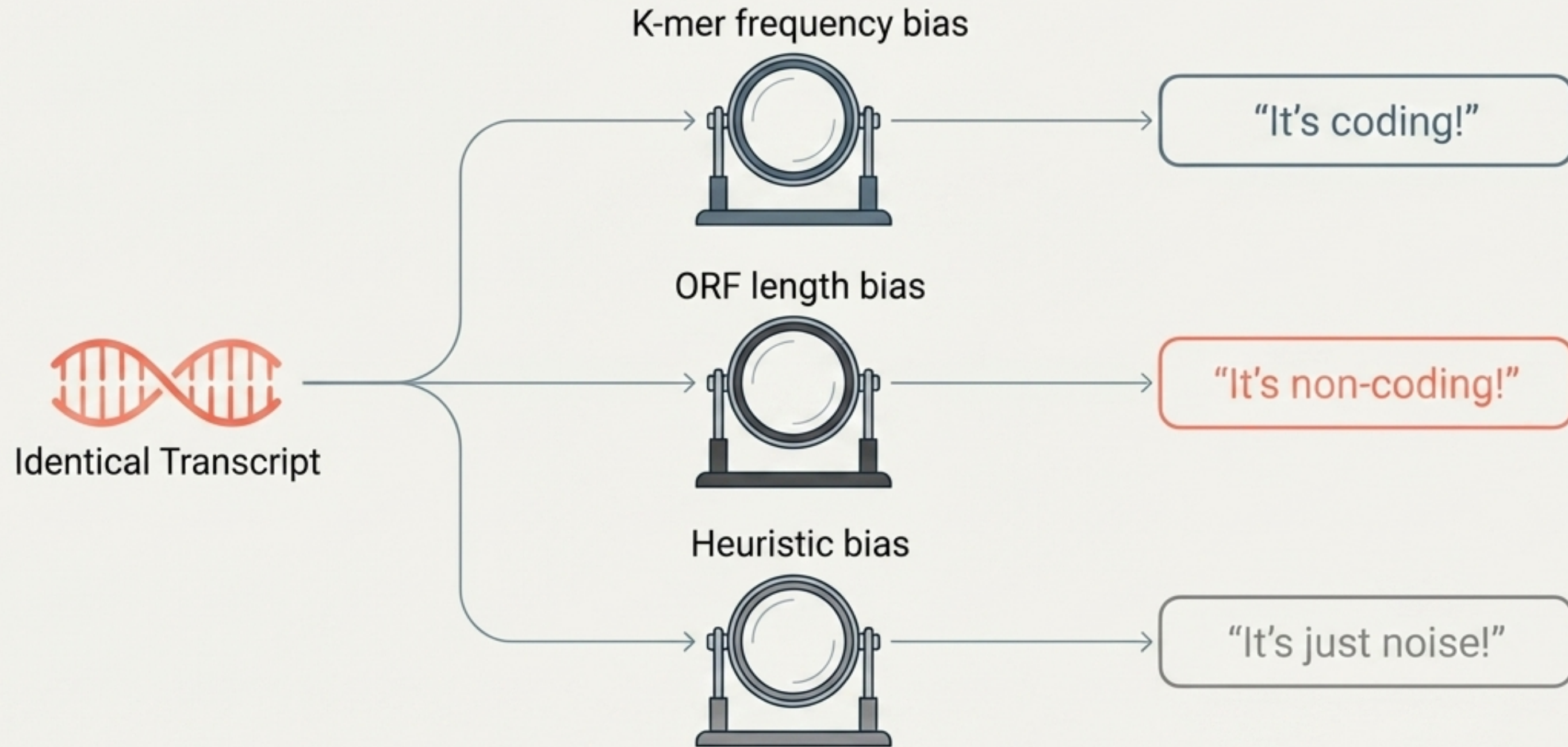
2. Missing Blueprints: They lack coding potential, making them nearly invisible to standard scanners.

3

The Result: They are easily confused with regular transcriptional noise or standard protein-coding sequences.

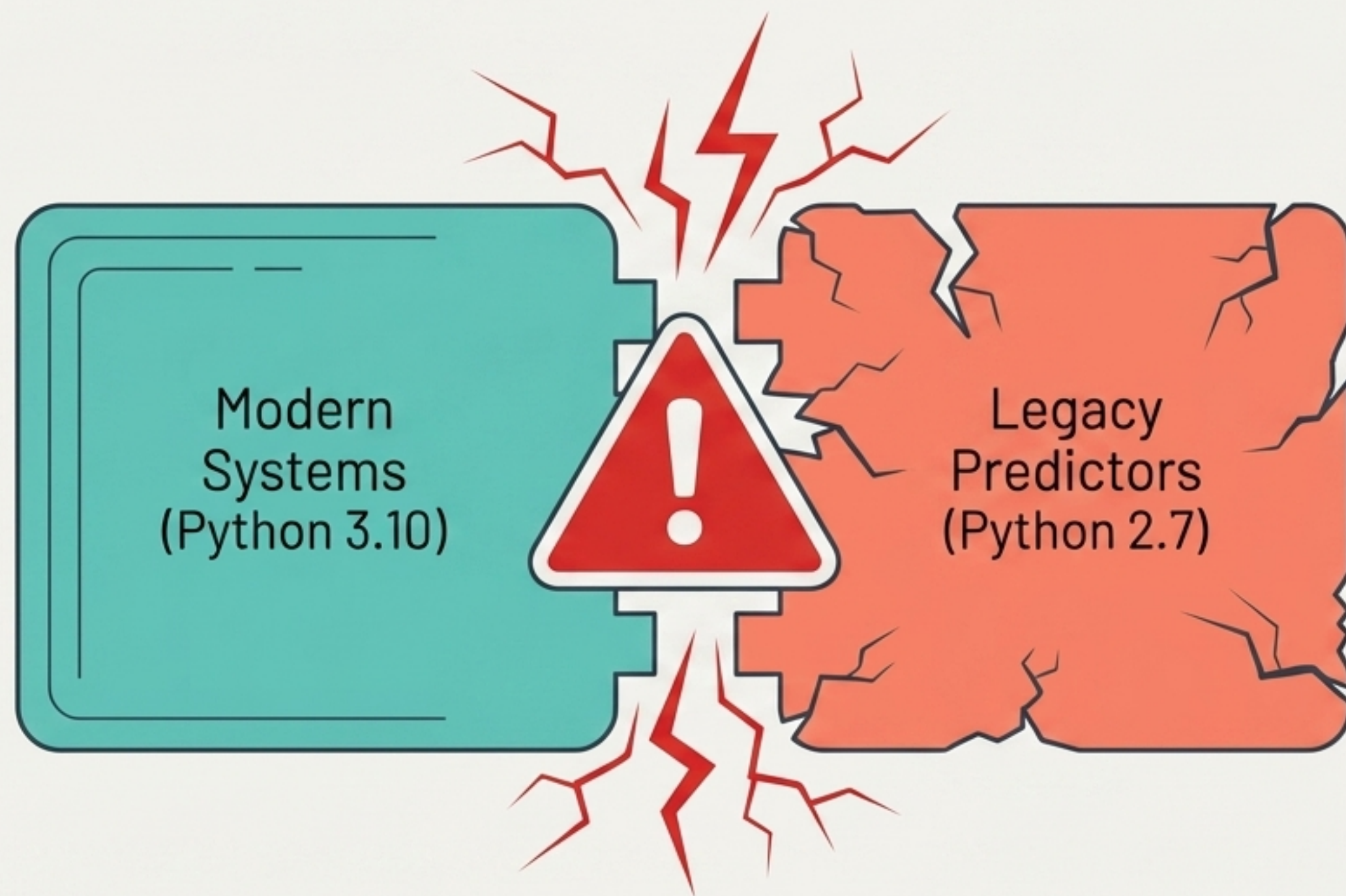


Roadblock 1: The Biased Magnifying Glasses



Methodological Instability: Individual predictive tools have low concordance rates. They are hyper-focused on specific features, leading to widely different answers when looking at the exact same transcript.

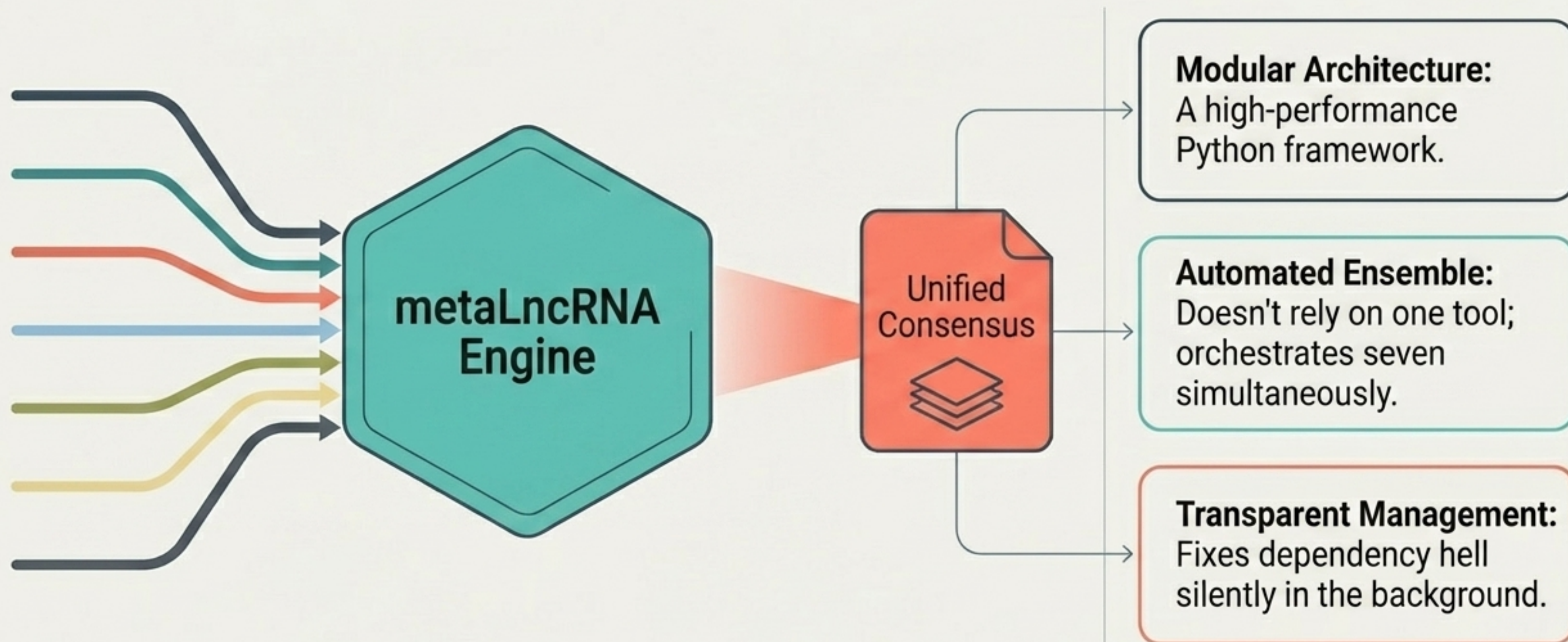
Roadblock 2: The Software Time Machine



The Reproducibility Gap

Many high-impact algorithms are abandoned by their creators. To run them today, scientists must downgrade their computers to use obsolete, conflicting software stacks—a frustrating nightmare known as Dependency Hell.

Enter metaLncRNA: The Master Coordinator



A Unified Board of 7 Robotic Experts

Statistical Codon Bias

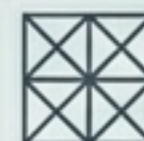


The Calculators

CPAT

LGC

Machine Learning



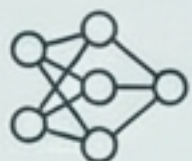
The Pattern Recognizers

CPC2

PLEK

CNCI

Deep Learning



The Neural Brain

RNAsamba

Integrative Features



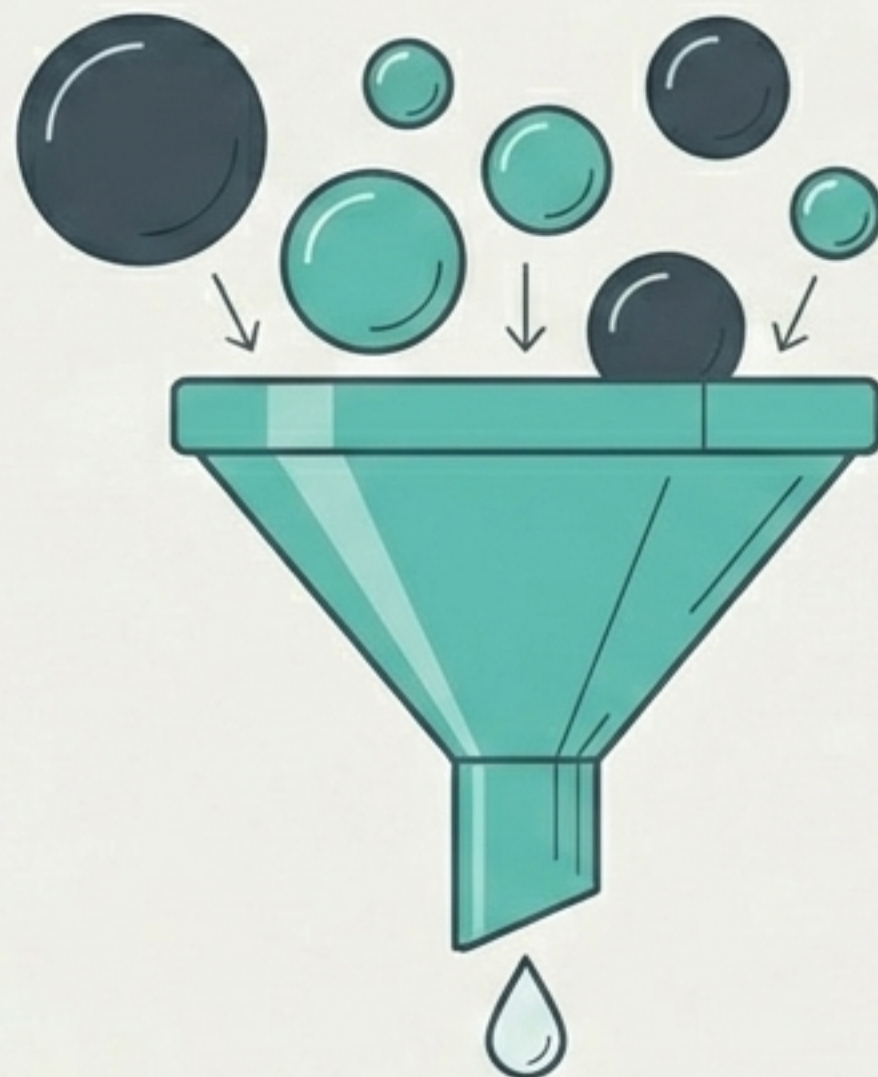
The Synthesizer

CPPred

No single expert is perfect, but together, they cover every possible angle of transcript analysis.

The Consensus Engine: Louder Voices for Better Accuracy

Historically more accurate tools are given a heavier “weight” in the final vote.

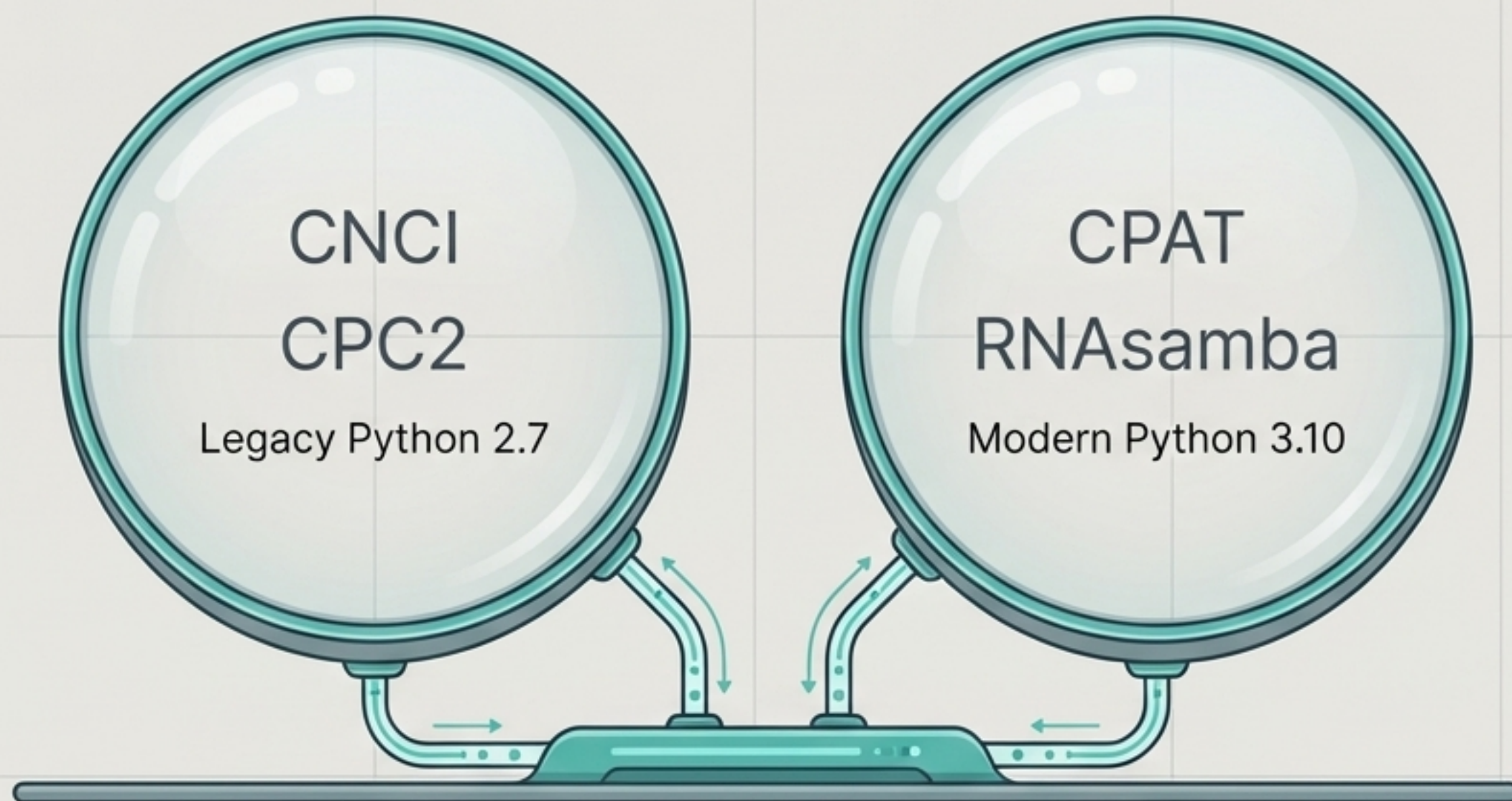


Meta-Score (S)

The Decision Rule

If $\text{Score} \leq 0.5$,
the transcript is
officially certified
as an **IncRNA**.

The Isolation Facility: Fixing Dependency Hell



Environment Orchestration

The 'metalncrna setup' command automatically builds isolated, sandboxed environments. Old and new software live in perfect harmony, guaranteeing long-term reproducibility without breaking your computer.

The AI Interpreter: From Numbers to Natural Language

The Old Way

0.842	0.113	0.002	0.002
0.842	0.113	0.002	0.567
0.113	0.113	0.905	0.002
0.905	0.842	0.113	0.002
0.902	0.113	0.905	0.002
CNCI	CPC2	CPAT	CPAT
0.902	0.902	0.113	0.002
0.942	0.113	0.913	0.002
0.113	0.905	0.905	0.002
0.313	0.905	0.067	0.003
0.905	0.132	0.113	0.002
0.905	0.902	0.905	0.002
0.313	0.113	0.905	0.002
0.005	0.123	0.903	0.002
0.113	0.113	0.902	0.002
0.842	0.113	0.867	0.013
0.906	0.905	0.002	0.016
0.927	0.102	0.002	0.002
0.878	0.113	0.905	0.002
0.324	0.905	0.902	0.002

The metaLncRNA Way

Tell me more about transcript_042.

Transcript 042 was classified as non-coding because despite PLEK's disagreement, the deep learning analysis found... **lncRNA**.



Total Data Privacy. The AI engine runs 100% locally on your machine. No genomic data is ever sent to the cloud.

Choose Your Dedicated AI Assistant



The Generalist (Llama-3.2)

Role: Standard summarization and basic explanations.

Hardware: Runs easily on standard laptops (Minimum 4GB RAM).

Best for: Everyday analysis and quick checks.



The Specialist (OpenBioLLM-8B)

Role: Deep biological context, fine-tuned on PubMed. Recommended for scientific publications.

Hardware: Requires heavier machinery (Minimum 8GB RAM, 16GB recommended).

Best for: Deep interpretation and resolving complex consensus disputes.

What You Actually Get



metalncrna_results.tsv

The Table: A unified consensus table with every individual score from the 7 experts and the final certified Meta-Score.



predicted_Incrnas.fasta

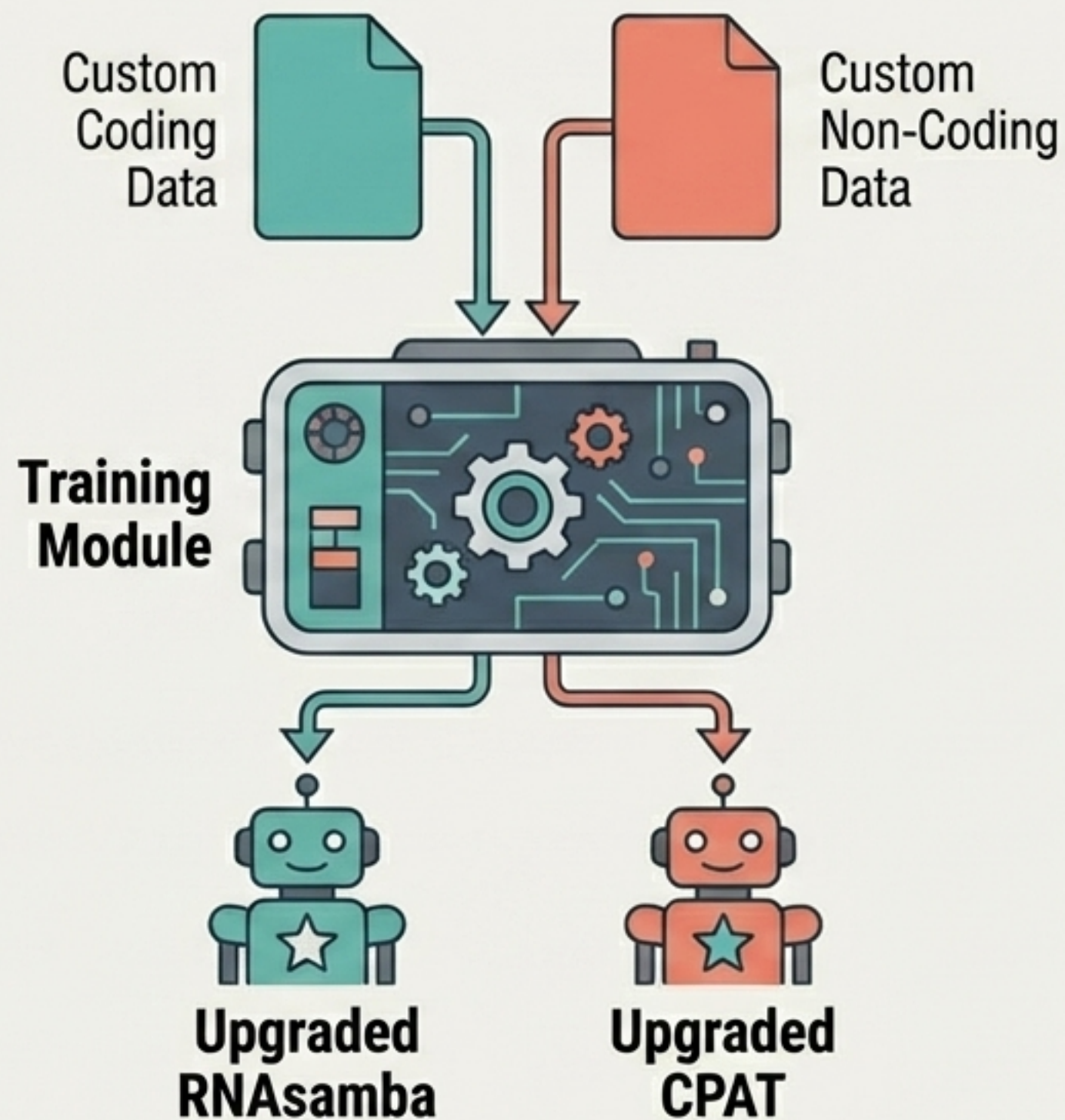
The Filtered Data: A pure, isolated file containing only the confirmed lncRNAs, ready for downstream biological analysis.



metalncrna.log

The Audit: A comprehensive execution log proving exactly how the system arrived at its conclusions, ensuring total traceability.

Studying a Brand New Species? Train Your Own Detective.

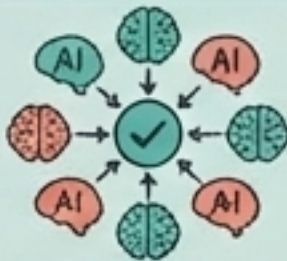
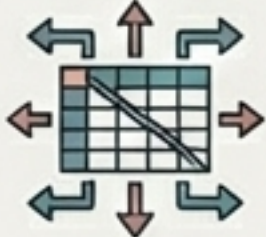


Extensibility is Built-In

Through the '**metalncrna train**' command, researchers working on non-model organisms can feed their own verified FASTA files to generate custom, species-specific models.

It transforms a simple predictor into a limitless research platform.

The Paradigm Shift

	Traditional Bioinformatics	The metaLncRNA Way
Decision Making	 One biased opinion	 7-expert weighted consensus
Software Stability	 Frequent crashes (dependency hell)	 Guaranteed isolation via Mamba
Outputs	 Confusing raw data tables	 Plain-English AI conversational insights
Resiliency	 Restarts from scratch if interrupted	 Checkpoint-aware (resumes exactly where it left off)

High-Powered Science, Zero Setup Headaches

Python & Bioconda

```
pip install
```

Standard, easy installation for everyday use.

Pixi

Modern, highly reproducible declarative development environments.

Docker & Singularity

Pre-configured containers for absolute reproducibility on massive supercomputers and HPCs.

Galaxy

A user-friendly, web-based platform. Click-and-play interface for experimental biologists.

A sustainable, reproducible future for genomic discoveries.